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Module: Mobile development
Lab 5: Posts Manager with SQLite

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1) Which dependencies did you use and why?

* Sqflite: the main sqflite plugin for flutter, it provides sqflite database engine that runs on the device for local data storage.
* Image picker: Allows people to pick images from gallery.
* path: used with sqflite to build the correct file path

Why sqlite

→ SQLite is serverless, self-contained database that runs directly on the device, it needs no internet connection, Unlike shared Preferences (Key-Value-only) SQL supports structured tables, complex queries and full CRUD operations, ideal for storing post offline.

2) How do you Handle database exceptions

- Database not initialized: the database getter uses lazy initialization, if openDatabase fails, the error is caught and rethrown with a ~~clear~~ clear message. "failed to initialize database".
- Insert/Update/Delete errors: Each CRUD method is wrapped in try-catch. if an operation fails (disk full, constraint error) the exception is caught and shown to the user via a SnackBar

- Invalid or corrupted data : The post. from map() factory uses null-safe casting with fall back defaults (?? ''), so the app won't crash even if a row has null or unexpected values
- Validating before insertion.

3) SQlite in flutter

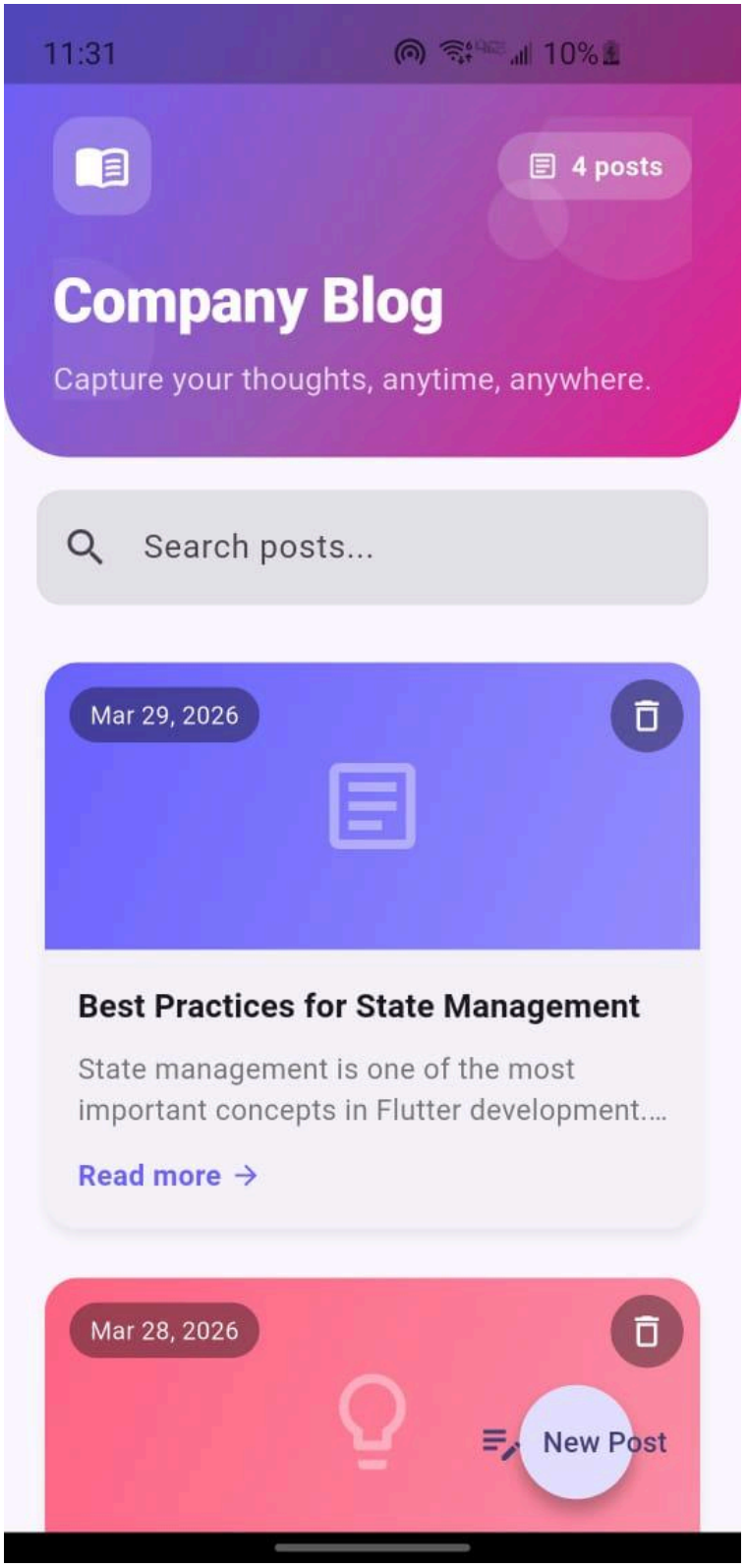
Database	Table
A database is a single file .db, stores all data	Collection of rows inside the database ex: post table id | title | description

CRUD operations

operation	Method
Create	db.insert('posts', post.toMap())
Read	db.query('posts')
Update	db.update('posts', post.toMap(), where: 'id' = ?')
Delete	db.delete('posts', where: 'id = ?')

Q How flutter interacts asynchronously

All sqlite methods returns futures, flutter uses async/await to run them without blocking the UI.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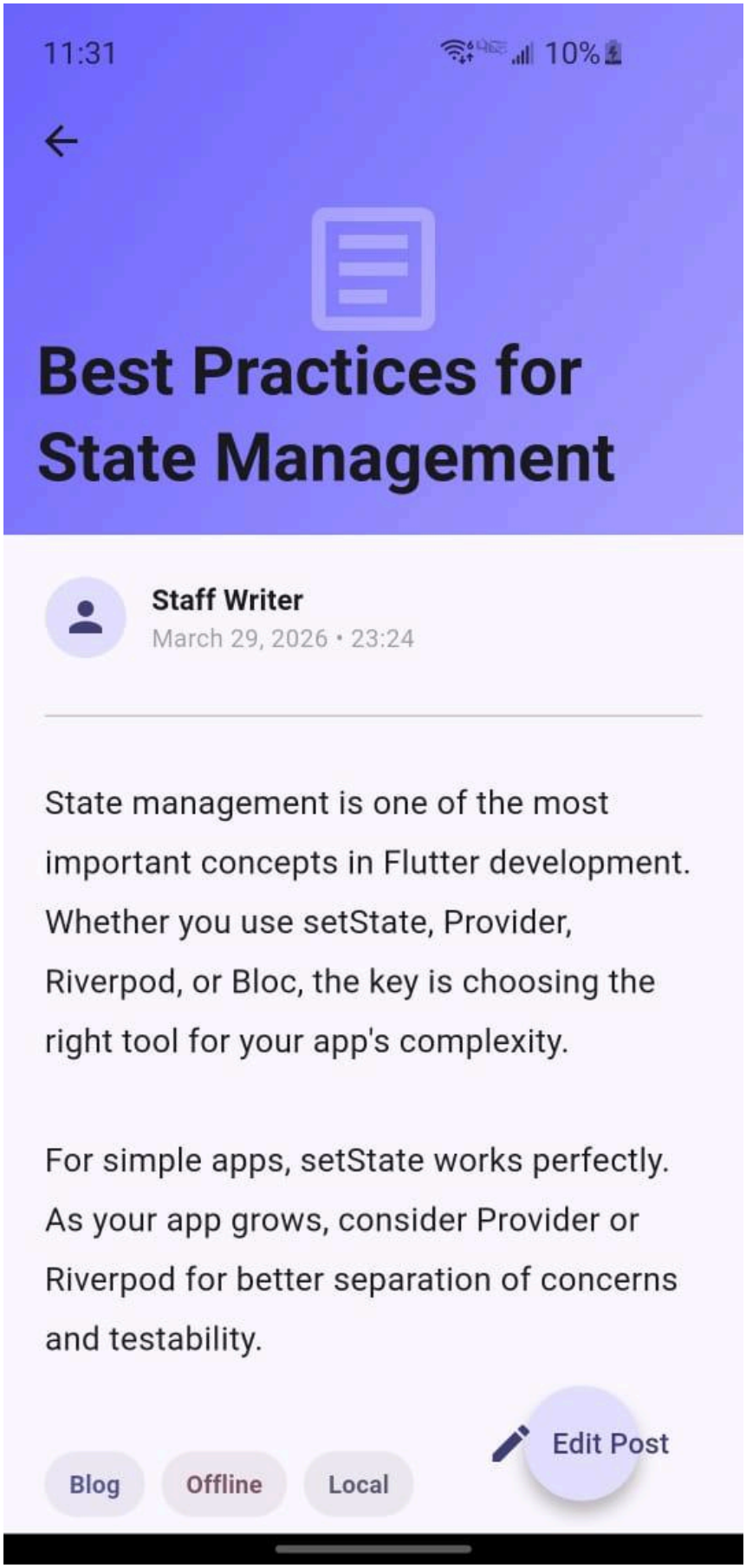
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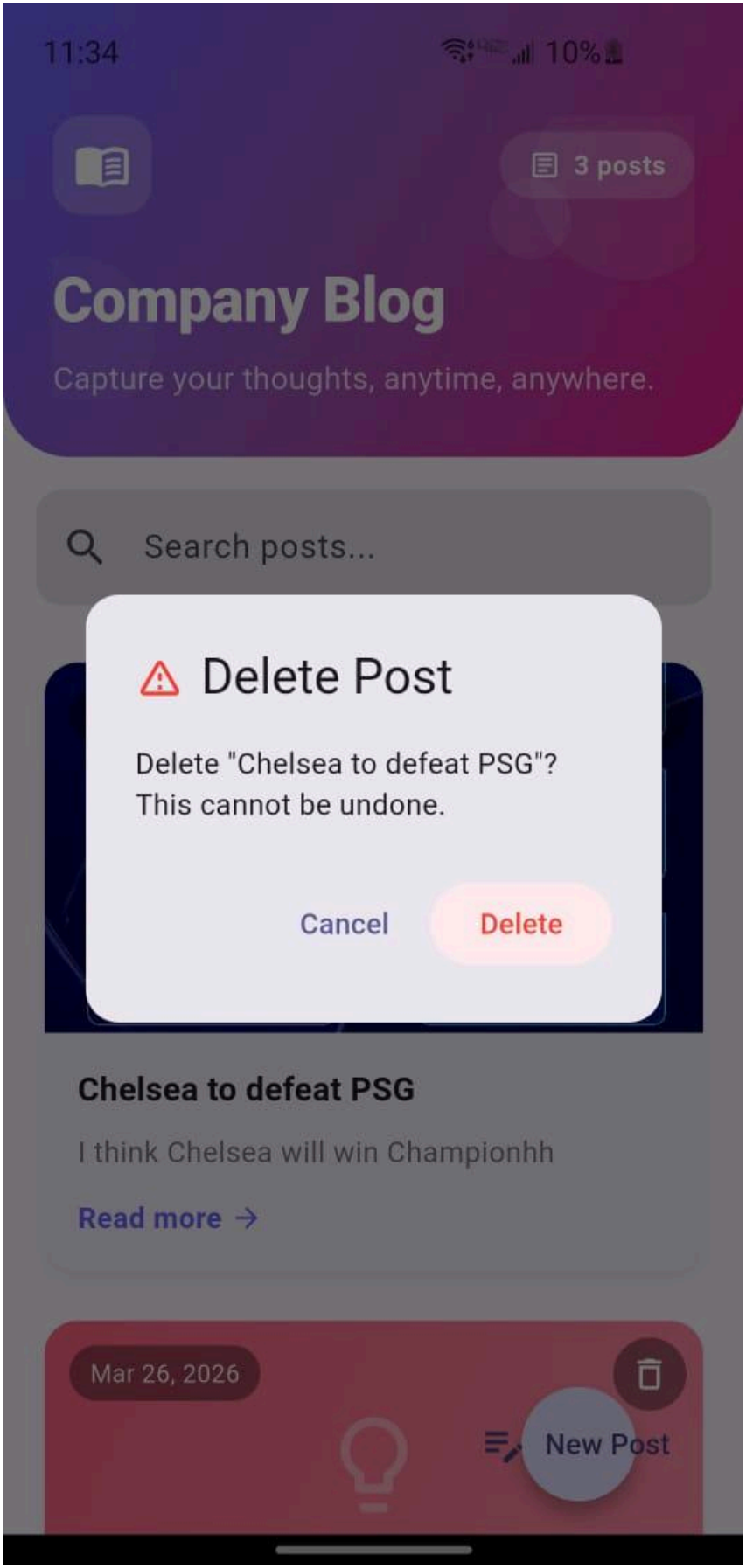
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